Week 7: Dimensionality Reduction

Feature Selection, PCA and Variance Visualisation on Road-Accident Data

DSV Team 14 | Source notebook: DSV-TEAM-14-WEEK-7.ipynb

1. Executive Summary

This report summarises the Week 7 notebook, which applies feature selection and Principal Component Analysis (PCA) to a road-accident dataset. Three numeric features, **Number_of_Vehicles**, **Number_of_Casualties** and **Speed_limit**, were screened for variance and for association with accident severity, standardised, and then projected with PCA.

Key results

- All three features vary across records (none were removed by the variance filter).
- Against accident severity, **Number_of_Vehicles** has the strongest ANOVA F-score (361.6), followed by **Number_of_Casualties** (248.3) and **Speed_limit** (202.6). All p-values are far below 0.001.
- PCA on the three features gave components explaining 43.4%, 33.1% and 23.5% of variance. The 90% target therefore required all three components, so no dimensionality was actually reduced.
- The 2-D projection shows heavy overlap between severity classes, so these three features do not separate severity on their own.

Important data-quality finding. The loaded file has 1,048,575 rows, but only 59,998 contain data; the other 988,577 are blank. The notebook imputed these blanks with medians before scaling and PCA, which distorts the standardisation and the PCA outputs (Section 5). The F-scores and variance chart are unaffected, but the PCA percentages and loadings should be re-run on the cleaned rows before being relied on.

2. Dataset and Objectives

The notebook loads *AccidentsBig.csv* (30 columns). It is a road-accident table with fields such as location, police force, severity, number of vehicles and casualties, road class, speed limit and junction details. The objectives of the week are to select suitable numeric features, standardise them, evaluate their relationship with severity, and apply PCA with explained-variance visualisations.

Item	Value
File loaded	AccidentsBig.csv
Shape as loaded	1,048,575 rows x 30 columns
Rows with data	59,998 (all other rows are entirely blank)
Exact duplicate rows / duplicate Accident_Index	988,576 each (the blank rows)
Total missing cells	29,657,408
Severity classes present	1, 2 and 3 (plus missing)

3. Methodology

Step	Method	Purpose
Feature choice	Number_of_Vehicles, Number_of_Casualties, Speed_limit	Continuous-style measurements; IDs and coded categories excluded
Imputation	SimpleImputer (median)	Fill missing numeric values
Scaling	StandardScaler	Put features on comparable scales for PCA
Variance filter	VarianceThreshold (0.0)	Remove constant features
Severity screening	SelectKBest, ANOVA F-test	Rank features by association with severity
PCA	scikit-learn PCA, 90% variance target	Create components and measure explained variance

4. Results: Feature Selection

Descriptive statistics for the 59,998 populated records:

Feature	Mean	Std dev	Min	Median	Max
Number_of_Vehicles	1.83	0.69	1	2	18
Number_of_Casualties	1.32	0.76	1	1	23
Speed_limit	35.06	11.46	10	30	70

Variance filter. No feature was constant, so all three were retained. The raw-variance chart shows Speed_limit (variance about 131) dwarfing the two count features (variance below 1). This is the reason standardisation is needed before PCA; otherwise Speed_limit would dominate the first component purely because of its units.

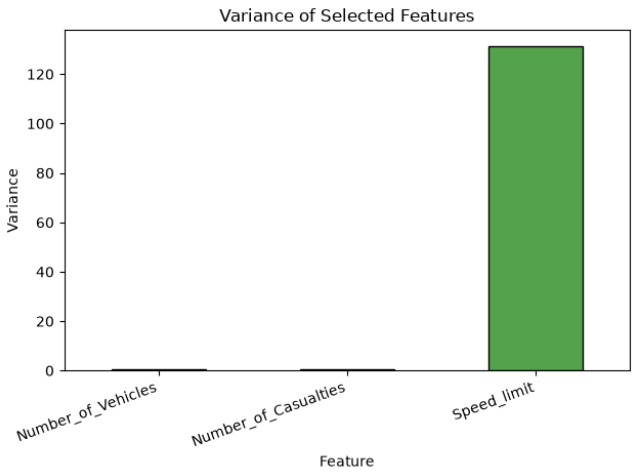


Figure 1. Variance of the selected features (raw units).

ANOVA F-scores against Accident_Severity.

Feature	F-score	p-value
Number_of_Vehicles	361.60	7.9e-157
Number_of_Casualties	248.27	4.2e-108
Speed_limit	202.63	2.0e-88

All three features differ significantly across severity classes. With about 60,000 records even small differences are statistically significant, so the F-scores are best used to rank features rather than to judge practical importance. The test indicates association only, not causation.



Figure 2. ANOVA F-scores by feature.

5. Results: PCA

Component	Explained variance (%)	Cumulative (%)
PC1	43.389	43.389
PC2	33.121	76.511
PC3	23.489	100.000

The variance is spread fairly evenly across the components. Two components reach only 76.5%, below both the 80% and 90% guide lines, so the notebook correctly selected all three components. With three input features this means the PCA is a rotation of the data rather than a reduction.

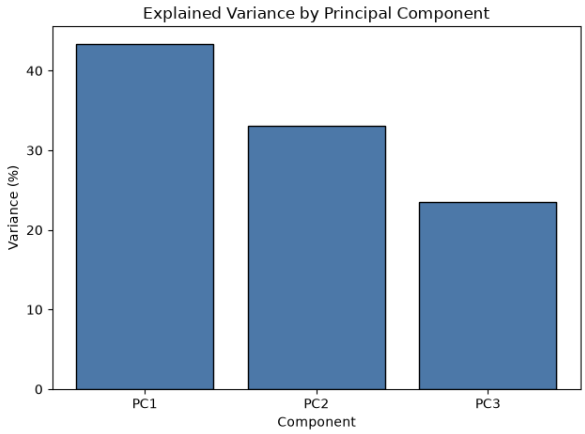


Figure 3. Explained variance by component.

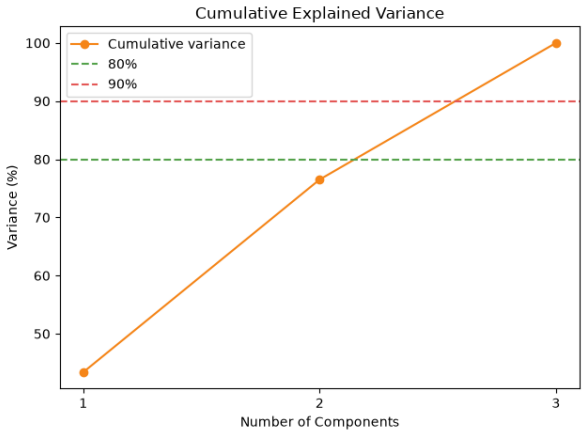


Figure 4. Cumulative explained variance.

Feature loadings

Feature	PC1	PC2	PC3
Number_of_Vehicles	0.2957	0.9172	-0.2669
Number_of_Casualties	0.7032	-0.0199	0.7107
Speed_limit	0.6466	-0.3978	-0.6509

PC1 is a broad combination led by casualties and speed limit. PC2 is almost entirely Number_of_Vehicles. PC3 contrasts casualties against speed limit. Signs can flip without changing the interpretation.

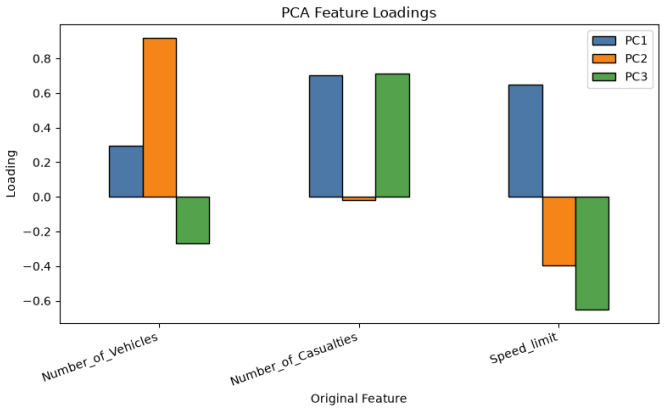


Figure 5. PCA feature loadings.

Two-dimensional projection

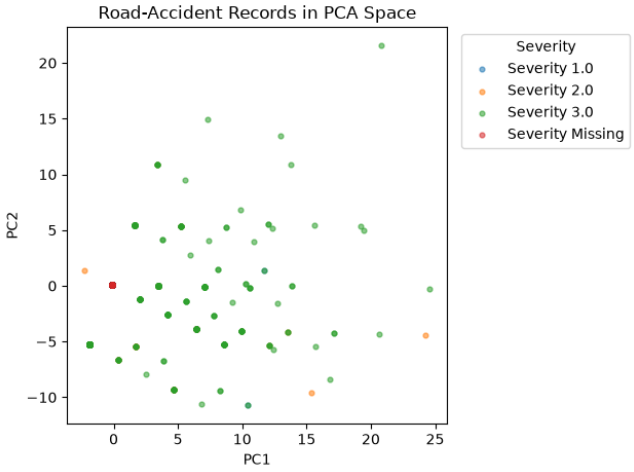


Figure 6. Records in PC1/PC2 space, coloured by severity (12,000-record sample).

Severity 3 points dominate and sit over the same regions as the rarer classes, so the first two components do not separate severity. A red "Missing" cluster is stacked at the origin; these are the blank rows described below.

6. Data-Quality Issue Affecting the PCA

> The notebook reports "Duplicate Accident_Index: 988,576" and 29.7 million missing cells. Those figures come from 988,577 entirely blank rows at the end of the CSV (the file stops at 1,048,575 rows, the spreadsheet row limit). They are not real accidents.

Because median imputation was applied to the whole table, about 94% of the matrix became identical median rows. This has three consequences:

- The standard deviation used for scaling collapses. In the output, a real record with one vehicle is scaled to about -5.8 instead of about -1.2.
- The PCA is fitted mostly on one repeated point, so the explained-variance split (43/33/23) and the loadings describe the distortion as much as the real data.
- The projection plot is crowded by the imputed cluster at the origin, with real records pushed to the edges.

Not affected: the descriptive statistics, raw variance chart and ANOVA F-scores, because those were computed on the 59,998 populated rows only.

7. Conclusions and Recommendations

The notebook demonstrates a complete, well-structured workflow for variance screening, ANOVA feature ranking, PCA and visual diagnostics. The ranking of features by severity association is sound: vehicles, then casualties, then speed limit. The PCA section needs one correction before its numbers are used.

- **Drop blank rows first.** Add `df = df.dropna(how="all")` (or `dropna(subset=["Accident_Index"])`) immediately after loading, then re-run Sections 3 to 15.
- **Expand the feature set.** Three features cannot be meaningfully reduced. Adding more numeric measurements (for example Time of day as hour, Day_of_Week, or road-related counts) would give PCA something to compress.

- **Handle severity imbalance.** Severity 3 dominates; consider class-aware evaluation or stratified sampling in later modelling weeks.
- **Treat coded categories carefully.** Fields such as Road_Type or Junction_Detail are codes, not magnitudes, and should be one-hot encoded rather than fed into PCA directly.
- **Save the reduced output** only after the re-run, using the optional step in Section 16.